Dwight Look College of ENGINEERING
TEXAS A&M UNIVERSITY

Team 56: OpenROAD VLSI Design 2

Bi-Weekly Update 5

Kevin, Talha, Githin, Giovani

Sponsor: Kevin Nowka

TA: Michael Johnson-Moore

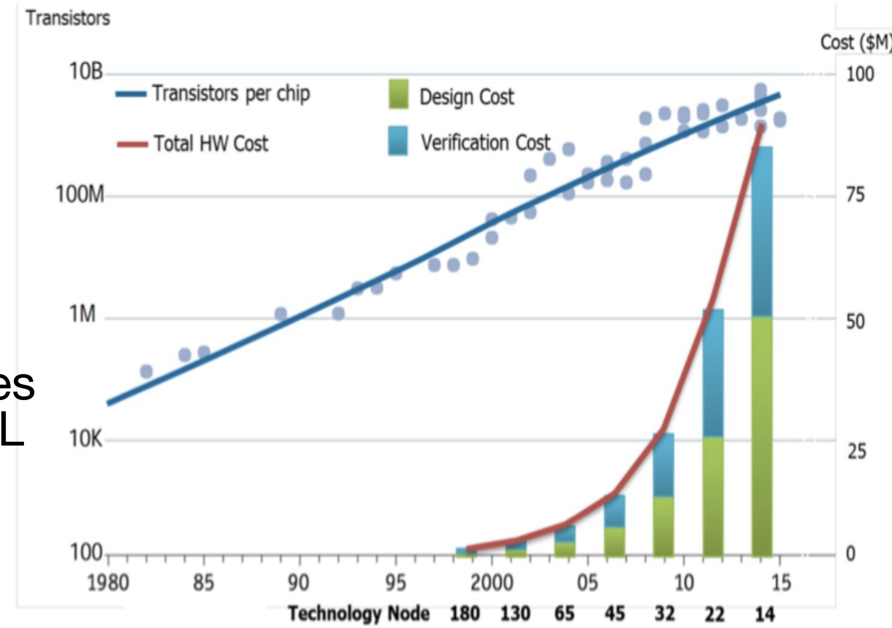
Project Summary

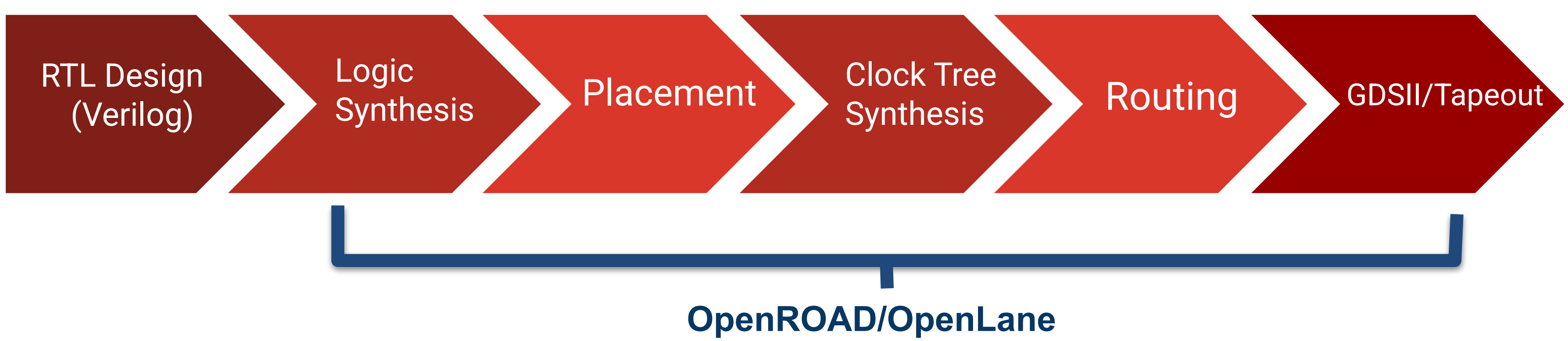
Problem statement:

- Traditional RTL-to-GDSII chip design is expensive, creating a cost barrier to developing a System on a Chip (SoC).

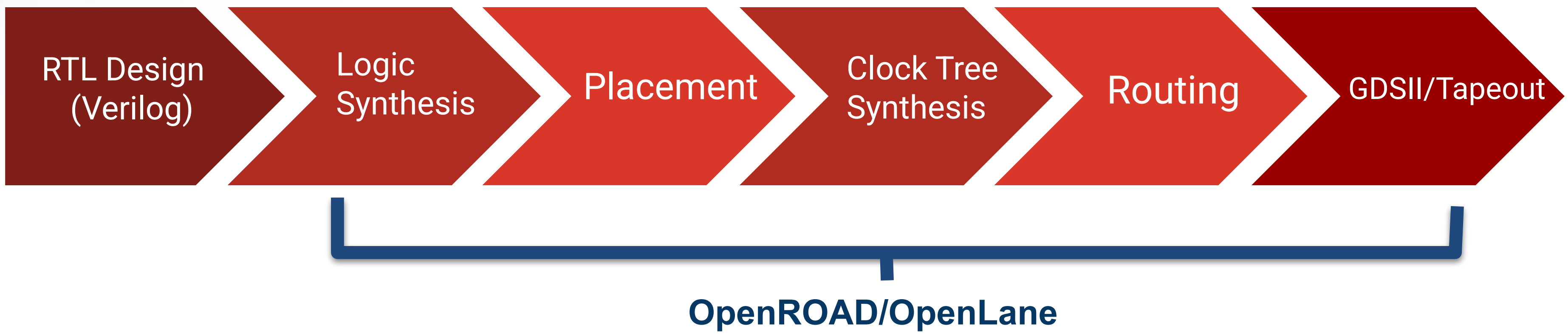
Our Solution:

- Create a 16-bit RISC-V using OpenROAD/OpenLane, which automates chip design by only needing only an RTL description to generate GDSII, significantly reducing cost.

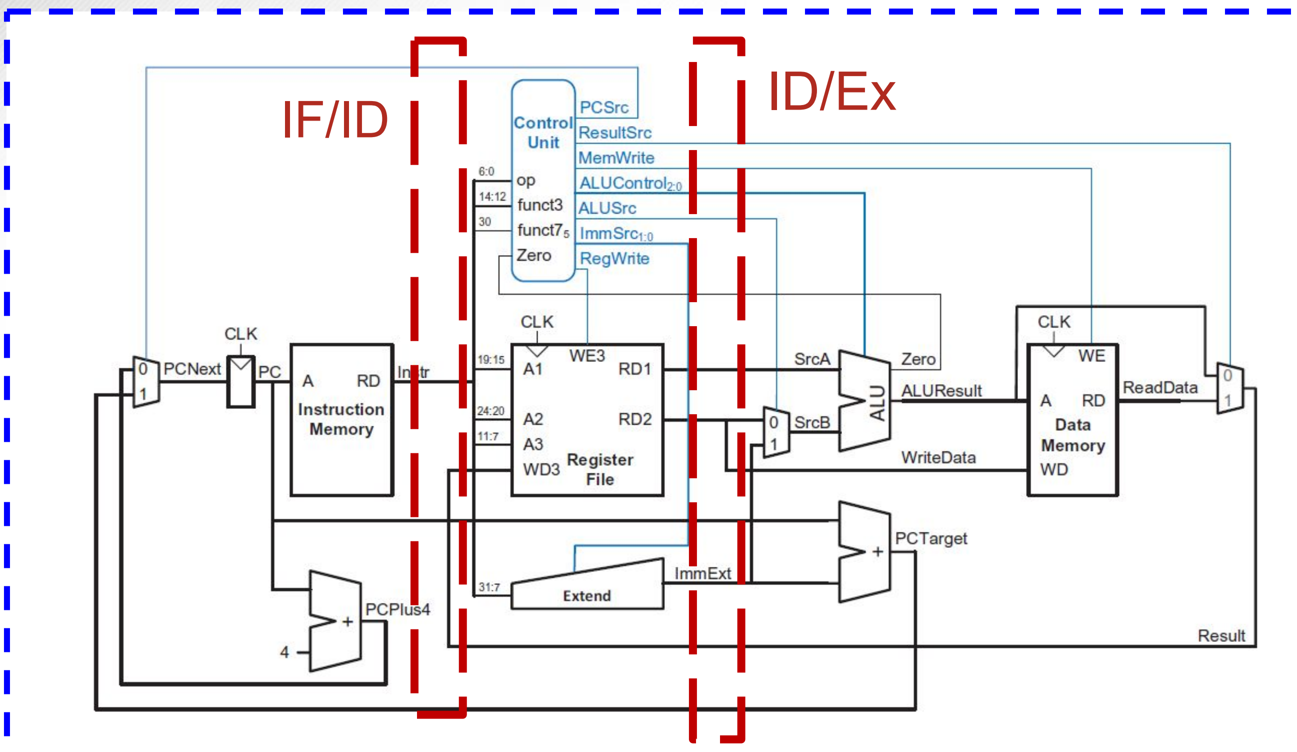




Design Flow


RTL Design (Verilog)
Logic Synthesis
Placement
Clock Tree Synthesis
Routing
GDSII/Tapeout
OpenROAD/OpenLane

System Overview



General Purpose Input Pins

General General Output Pins

Project Timeline

Finalize individual subsystem (completed 1/26)	Start integration/validation (Start by 2/2)	Implement pipeline stages & GPIO (Start by 2/16)	Validate pipeline stages & GPIO (Start by 3/2)	Start OpenLane/OpenROAD process (Start by 3/23)	Final Validation (Start by 3/23)

3 Stage Pipeline

Githin Johny & Giovani Giagnacovo

Accomplishments since status update 4 11 hrs of effort	Ongoing progress/problems and plans until next presentation
• Successfully got all test cases for each instruction type to pass for 3 stage pipeline implementation	• Continue validating pipelined design to test limits more comprehensively • Work on optimizing finished CPU design for fabrication

Pipeline Testbench Summary

```
================================================================
  RISC-V16 Pipeline - Full Instruction Type Coverage
================================================================

--- I-type: ADDI ---
  [PASS] ADDI x1 = 3                            got=0x0003
  [PASS] ADDI x2 = 2                            got=0x0002

--- R-type: ADD / SUB / AND / OR / XOR ---
  [FAIL] ADD  x3 = 5  (3+2)                     got=0x0000  exp=0x0005
  [PASS] SUB  x4 = 1  (3-2)                     got=0x0001
  [PASS] AND  x5 = 2  (3&2)                     got=0x0002
  [PASS] XOR  x7 = 1  (3^2)                     got=0x0001

--- I-type: ANDI/ORI/XORI then JALR link stored in x3 ---
  [PASS] JALR link x3 = 0x0026                  got=0x0026

--- S-type: SW  /  L-type: LW  (load-use stall) ---
  [PASS] LW x5 = 2  (mem[0])                    got=0x0002

--- B-type: BEQ taken (flush kills ADDI x1,-1) ---
  [PASS] BEQ flush: x1 still 3                  got=0x0003
  [PASS] BNE flush: x2 still 2                  got=0x0002

--- J-type: JAL ---
  [PASS] JAL link x6 = 0x0022                   got=0x0022
  [PASS] JAL flush: x1 still 3                  got=0x0003

--- JR-type: JALR x3, x6, 0 ---
  [PASS] JALR link x3 = 0x0026                  got=0x0026

================================================================
  Results: 12 PASSED,  1 FAILED
================================================================
```

```
45000  | PC=0000 instr=0000 ALU=0000 | x1=0000 x2=0000 x3=0000 x6=0000
55000  | PC=0000 instr=6081 ALU=0003 | x1=0000 x2=0000 x3=0000 x6=0000
65000  | PC=0002 instr=4101 ALU=0002 | x1=0000 x2=0000 x3=0000 x6=0000
75000  | PC=0004 instr=2800 ALU=0005 | x1=0000 x2=0000 x3=0000 x6=0000
85000  | PC=0006 instr=2a08 ALU=0001 | x1=0003 x2=0000 x3=0000 x6=0000
95000  | PC=0008 instr=2ac0 ALU=0000 | x1=0003 x2=0002 x3=0000 x6=0000
105000 | PC=000a instr=2b18 ALU=0003 | x1=0003 x2=0002 x3=0000 x6=0000
115000 | PC=000c instr=2b90 ALU=0001 | x1=0003 x2=0002 x3=0000 x6=0000
125000 | PC=000e instr=6199 ALU=0000 | x1=0003 x2=0002 x3=0000 x6=0000
135000 | PC=0010 instr=4191 ALU=0002 | x1=0003 x2=0002 x3=0000 x6=0003
145000 | PC=0012 instr=2189 ALU=0003 | x1=0003 x2=0002 x3=0000 x6=0003
155000 | PC=0014 instr=0013 ALU=0000 | x1=0003 x2=0002 x3=0000 x6=0003
165000 | PC=0016 instr=002a ALU=0000 | x1=0003 x2=0002 x3=0002 x6=0003
175000 | PC=0018 instr=12a4 ALU=0002 | x1=0003 x2=0002 x3=0003 x6=0003
185000 | PC=0018 instr=12a4 ALU=0000 | x1=0003 x2=0002 x3=0003 x6=0003
195000 | PC=0000 instr=0000 ALU=0000 | x1=0003 x2=0002 x3=0003 x6=0003
205000 | PC=001c instr=114c ALU=0001 | x1=0003 x2=0002 x3=0003 x6=0003
215000 | PC=0000 instr=0000 ALU=0000 | x1=0003 x2=0002 x3=0003 x6=0003
225000 | PC=0020 instr=00b5 ALU=0002 | x1=0003 x2=0002 x3=0003 x6=0003
235000 | PC=0000 instr=0000 ALU=0000 | x1=0003 x2=0002 x3=0003 x6=0003
245000 | PC=0024 instr=00df ALU=0022 | x1=0003 x2=0002 x3=0003 x6=0003
255000 | PC=0000 instr=0000 ALU=0000 | x1=0003 x2=0002 x3=0003 x6=0022
265000 | PC=0022 instr=0000 ALU=0000 | x1=0003 x2=0002 x3=0003 x6=0022
275000 | PC=0024 instr=00df ALU=0022 | x1=0003 x2=0002 x3=0026 x6=0022
285000 | PC=0000 instr=0000 ALU=0000 | x1=0003 x2=0002 x3=0026 x6=0022
295000 | PC=0022 instr=0000 ALU=0000 | x1=0003 x2=0002 x3=0026 x6=0022
305000 | PC=0024 instr=00df ALU=0022 | x1=0003 x2=0002 x3=0026 x6=0022
315000 | PC=0000 instr=0000 ALU=0000 | x1=0003 x2=0002 x3=0026 x6=0022
325000 | PC=0022 instr=0000 ALU=0000 | x1=0003 x2=0002 x3=0026 x6=0022
335000 | PC=0024 instr=00df ALU=0022 | x1=0003 x2=0002 x3=0026 x6=0022
345000 | PC=0000 instr=0000 ALU=0000 | x1=0003 x2=0002 x3=0026 x6=0022
355000 | PC=0022 instr=0000 ALU=0000 | x1=0003 x2=0002 x3=0026 x6=0022
365000 | PC=0024 instr=00df ALU=0022 | x1=0003 x2=0002 x3=0026 x6=0022
375000 | PC=0000 instr=0000 ALU=0000 | x1=0003 x2=0002 x3=0026 x6=0022
385000 | PC=0022 instr=0000 ALU=0000 | x1=0003 x2=0002 x3=0026 x6=0022
```

GPIO & OpenLane

Kevin Umanzor & Talha Ibrahim

Accomplishments since status update 4 11 hrs of effort	Ongoing progress/problems and plans until next presentation
• Finished Validating Bootstrap FSM and IO ports • Successfully ran OpenLane & obtaining important metrics	• Working on optimizing design for project specifications

I/O Test

```
================================================
PHASE 1: LOAD MODE
================================================
  [LOAD] instr=0x2901  T1:ui=0x01 uio=0xa4  T2:ui[5:4]=0b00
  [LOAD] ACK OK  (addr=0)
  [LOAD] instr=0x2901  T1:ui=0x01 uio=0xa4  T2:ui[5:4]=0b00
  [LOAD] ACK OK  (addr=1)
  [LOAD] instr=0x2901  T1:ui=0x01 uio=0xa4  T2:ui[5:4]=0b00
  [LOAD] ACK OK  (addr=2)
  [LOAD] instr=0x3101  T1:ui=0x01 uio=0xc4  T2:ui[5:4]=0b00
  [LOAD] ACK OK  (addr=3)
  [LOAD] instr=0x0000  T1:ui=0x00 uio=0x00  T2:ui[5:4]=0b00
  [LOAD] ACK OK  (addr=4
  [LOAD] instr=0x0000  T
  [LOAD] ACK OK  (addr=5
```

```
================================================
PHASE 4: Register checks
================================================
  PASS | x1=0 untouched                        | got 0x0000
  PASS | x2=4 after ADDI x2,x4,1               | got 0x0004
  PASS | x3=0 untouched                        | got 0x0000

================================================
FINAL: pass=4 fail=0
================================================
```

Area Spec

```
Updating metrics…
Cell type report:                                  Count       Area
  Fill cell                                         1024    3843.69
  Tap cell                                         27756   34728.31
  Antenna cell                                     22137   55395.63
  Buffer                                               8      30.03
  Clock buffer                                      1120   14821.72
  Timing Repair Buffer                              6504   59583.40
  Inverter                                            34     135.13
  Clock inverter                                     462    6602.58
  Sequential cell                                   4503   91163.68
  Multi-Input combinational cell                    9139   94677.05
  Total                                            72687  360981.21
Writing OpenROAD database to '/Users/talhaibrahim/my_designs/CPU/ru
```

```
        "design__core__area": 1941700,
        "design__instance__count__stdcell": 71663,
        "design__instance__area__stdcell": 357138,
        "design__instance__count__macros": 0,
        "design__instance__area__macros": 0,
        "design__instance__utilization": 0.18393,
        "design__instance__utilization__stdcell": 0.18393,
        "design__instance__count__class:buffer": 8,
```

Die Area

```
[INFO IFP-0001] Added 512 rows of 3031 site unithd.
[INFO IFP-0030] Inserted 0 tiecells using sky130_fd_sc_hd__conb_1/LO.
[INFO IFP-0030] Inserted 0 tiecells using sky130_fd_sc_hd__conb_1/HI.
[INFO] Extracting DIE_AREA and CORE_AREA from the floorplan
[INFO] Floorplanned on a die area of 0.0 0.0 1405.36 1416.08 (µm).
[INFO] Floorplanned on a core area of 5.52 10.88 1399.78 1403.52 (µm).
Writing metric design__die__bbox: 0.0 0.0 1405.36 1416.08
Writing metric design__core__bbox: 5.52 10.88 1399.78 1403.52
Setting global connections for newly added cells…
```

STA

```
  Fanout         Cap        Slew       Delay        Time   Description
-------------------------------------------------------------------------------------
                        0.150000    0.000000    0.000000   clock clk (rise edge)
                                    0.000000    0.000000   clock network delay (ideal)
                                    8.000000    8.000000 ^ input external delay
     140    0.506271    1.698526    1.184126    9.184126 ^ rst_n (in)
                                                           rst_n (net)
                        1.698526    0.000000    9.184126 ^ _20475_/RESET_B (sky130_fd_sc_hd__dfrtp_2)
                                                9.184126   data arrival time

                        0.150000   40.000000   40.000000   clock clk (rise edge)
                                    0.000000   40.000000   clock network delay (ideal)
                                   -0.250000   39.750000   clock uncertainty
                                    0.000000   39.750000   clock reconvergence pessimism
                                               39.750000 ^ _20475_/CLK (sky130_fd_sc_hd__dfrtp_2)
                                   -0.308231   39.441769   library recovery time
                                               39.441769   data required time
-------------------------------------------------------------------------------------
                                               39.441769   data required time
                                               -9.184126   data arrival time
-------------------------------------------------------------------------------------
                                               30.257647   slack (MET)
```

DRC

```
    390000 uses
    395000 uses
    400000 uses
    405000 uses
    410000 uses
    415000 uses
[INFO] Loading tt_um_riscv16

DRC style is now "drc(full)"
Loading DRC CIF style.
No errors found.
[INFO] COUNT: 0
[INFO] Should be divided by 3 or 4
[INFO] DRC Checking DONE (/Users/talhaibrahim/my_designs/CPU/runs/RUN_2026-03-30_08-34-19/62-magic-drc/reports/drc_vio
[INFO] Saving mag view with DRC errors (/Users/talhaibrahim/my_designs/CPU/runs/RUN_2026-03-30_08-34-19/62-magic-drc/vi
[INFO] Saved
```

Dwight Look College of ENGINEERING TEXAS A&M UNIVERSITY

LVS

```
uio_in[0]                                    |uio_in[0]
uio_in[2]                                    |uio_in[2]
uio_in[7]                                    |uio_in[7]
clk                                          |clk
VGND                                         |VGND
VPWR                                         |VPWR
ena                                          |ena
---------------------------------------------------------------------------------------------
Cell pin lists are equivalent.
Device classes tt_um_riscv16 and tt_um_riscv16 are equivalent.

Final result: Circuits match uniquely.
.

```

Power

CPU > runs > RUN_2026-03-30_09-00-53 > 12-openroad-staprepnr > nom_tt_025C_1v80 > power.rpt

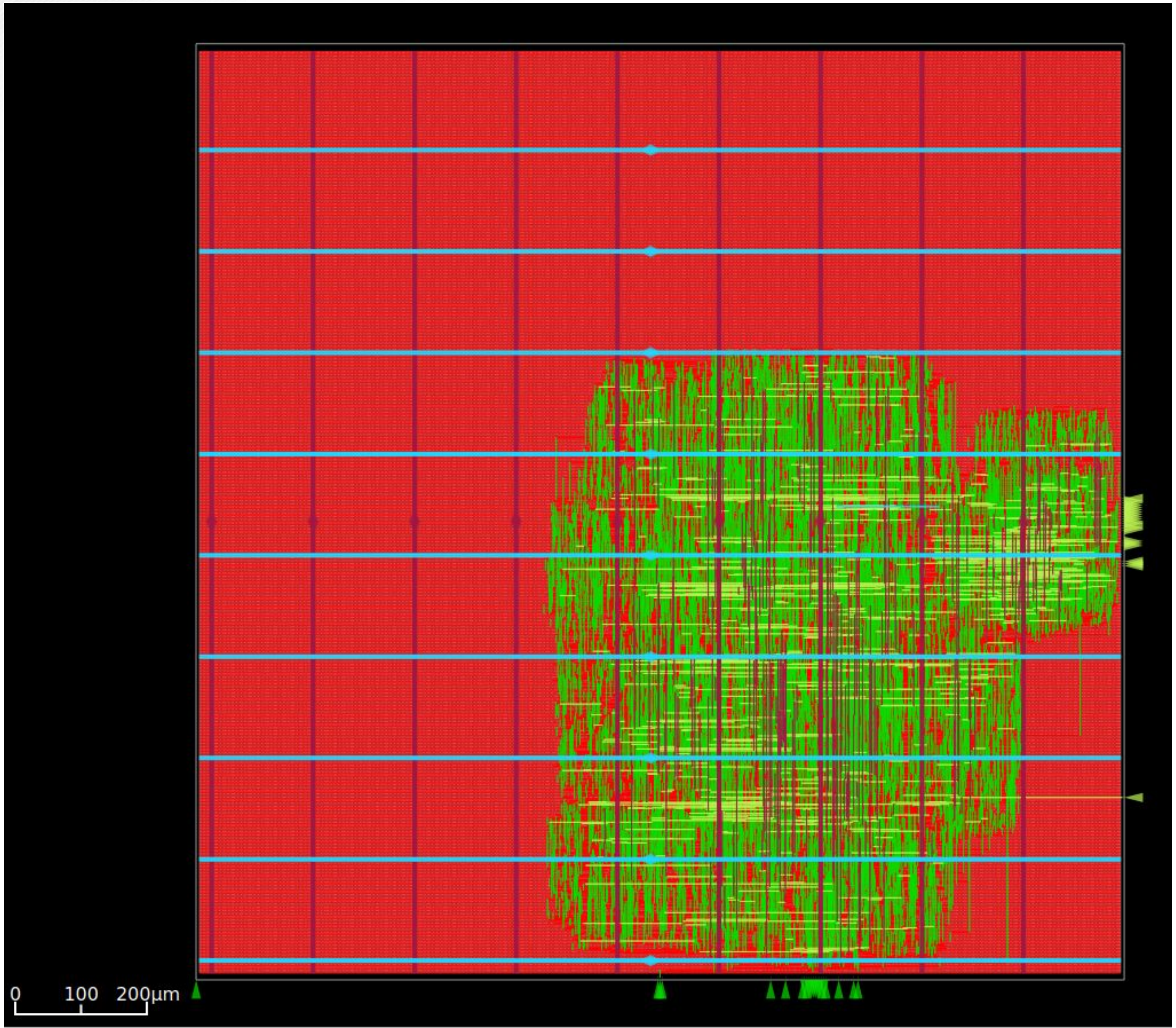
```

===========================================================================
 report_power
============================================================================
======================== nom_tt_025C_1v80 Corner ===================================

Group                  Internal  Switching    Leakage      Total
                          Power      Power      Power      Power (Watts)
----------------------------------------------------------------------------
Sequential             4.690573e-03 3.095727e-04 3.896356e-08 5.000184e-03  39.6%
Combinational          5.477844e-03 2.148002e-03 4.657001e-08 7.625893e-03  60.4%
Clock                  0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00   0.0%
Macro                  0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00   0.0%
Pad                    0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00   0.0%
----------------------------------------------------------------------------
Total                  1.016921e-02 2.457575e-03 8.553488e-08 1.262687e-02 100.0%
                              80.5%        19.5%         0.0%

```

Final GDSII Layout



Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

Execution Plan

Validation Plan

Paragraph #	Test Name	Success Criteria	Methodology	Status	Responsible Engineer(s)
3.1	Tile Size Constraint	The final GDSII layout must remain within the 2-tile maximum area allowed for TinyTapeout.	Run final Place and Route in OpenROAD, verify bounding box and padframe placement using the layout area report.	Tested	Full Team
4.1	DRC	Zero DRC violations (including antenna) in Magic and KLayout using the Sky130A/TinyTapeout rule decks.	Run Magic and KLayout DRC on the final GDS and iterate fixes until zero violations.	Tested	Full Team
4.2	Operating Temperature Range	Design must remain functionally valid across process and temperature corners (-55°C to +125°C).	Run timing and power analysis using PDK corner simulations (TT, SS, FF) to ensure performance remains stable within range.	Untested	Talha Ibrahim, Kevin Umanzor
5.1	LVS	Netgen reports “LVS clean” with 0 opens/shorts and all devices/nets/pins matched.	Extract the layout to SPICE and compare against the synthesized gate-level netlist in Netgen with the sky130A setup.	Tested	Full Team
5.2	Primary Input Power	Active power below 200 μW and lower during idle conditions.	Estimate and verify power through post-layout power analysis and simulation reports in OpenROAD.	Tested	Talha Ibrahim, Kevin Umanzor
5.3	Signal Interfaces	All interfaces (I^2C, SPI, USB) respond correctly at specified logic levels and timings.	Run RTL and gate-level testbenches to verify correct data transfer, logic levels, and signal timing for all interfaces.	Tested	Githin Johny, Giovani Giagnacovo
5.4	User Control Interface	The reset mechanism returns the CPU to a known initial state.	Run RTL simulations for both synchronous and asynchronous resets to confirm the PC and register file reset correctly.	Tested	Giovani Giagnacovo, Talha Ibrahim
5.5	Clock Interface	The design accepts a 10 MHz external clock and meets all setup and hold timing constraints.	Run Static Timing Analysis (STA) and post-layout timing simulations in OpenROAD to verify clock stability and timing closure.	Tested	Kevin Umanzor, Githin Johny
6.1	Fetch	The Fetch stage retrieves correct instructions and updates the program counter properly.	Create directed and random test programs in RTL simulation to verify instruction fetch and PC increment behavior.	Tested	Kevin Umanzor
6.2	Decode	The Decode stage correctly generates control signals from instruction fields.	Run RTL testbenches for all supported instructions to confirm control signal accuracy and complete coverage.	Tested	Githin Johny
6.3	Execute	The ALU produces correct results for all arithmetic and logic operations.	Run RTL and gate-level simulations for arithmetic, logical, and branch operations with directed and random test vectors.	Tested	Talha Ibrahim
6.4	Memory/Write Back	The memory and write-back stages perform correct data reads, writes, and register updates.	Run RTL and gate-level simulations with memory models to verify correct load/store behavior and data integrity.	Tested	Giovani Giagnacovo
	Full System Simulation	The full 16-bit RISC-V design executes test programs correctly through all pipeline stages.	Integrate all modules into a top-level RTL and post-layout simulation, verifying instruction execution, output correctness, and timing.	Tested	Full Team
	Full System Demo	Execute a test program on each corresponding part on the chip mesuring correct output for all cases; along with measured power and basic I/O pass	Integrate tests for top level regeression and postlayout simulation captruing outputs.	Untested	Full Team

Thank You!